

SOLUTIONS

1) Find the derivative with respect to x for each function.

a) $y = 4 \sin x$

$$y' = 4 \cos x$$

b) $f(x) = -3 \cos x$

$$f'(x) = -3(-\sin x)$$

$$f'(x) = 3 \sin x$$

c) $y = \cos x - \sin x$

$$y' = -\sin x - \cos x$$

d) $y = x^2 - 3 \sin x$

$$y' = 2x - 3 \cos x$$

e) $y = \cos x + 7\pi \sin x - 3x$

$$y' = -\sin x + 7\pi \cos x - 3$$

f) $f(x) = \frac{\pi}{4} \cos x - \frac{\pi}{3} \sin x$

$$f'(x) = \frac{\pi}{4}(-\sin x) - \frac{\pi}{3} \cos x$$

$$f'(x) = -\frac{\pi}{4} \sin x - \frac{\pi}{3} \cos x$$

Find the equation of the line that is tangent to the function $y = \cos x$ and passes through the point $(\frac{\pi}{3}, \frac{1}{2})$.

Slope:

$$y' = -\sin x$$

$$y'(\frac{\pi}{3}) = -\sin(\frac{\pi}{3})$$

$$y'(\frac{\pi}{3}) = -\frac{\sqrt{3}}{2}$$

$$m = -\frac{\sqrt{3}}{2}$$

Eqⁿ:

$$y = mx + b$$

$$\frac{1}{2} = (-\frac{\sqrt{3}}{2})(\frac{\pi}{3}) + b$$

$$\frac{1}{2} = -\frac{\sqrt{3}\pi}{6} + b$$

$$\frac{3}{6} + \frac{\sqrt{3}\pi}{6} = b$$

$$b = \frac{3 + \sqrt{3}\pi}{6}$$

$$y = -\frac{\sqrt{3}}{2}x + \frac{3 + \pi\sqrt{3}}{6}$$

3) Find the equation of the line that is tangent to the function $y = -4 \sin x$ at $x = \frac{\pi}{4}$.

Point:

$$y(\frac{\pi}{4}) = -4 \sin(\frac{\pi}{4})$$

$$y(\frac{\pi}{4}) = -4(\frac{1}{\sqrt{2}})$$

$$y(\frac{\pi}{4}) = -4(\frac{\sqrt{2}}{2})$$

$$y(\frac{\pi}{4}) = -2\sqrt{2}$$

$$(\frac{\pi}{4}, -2\sqrt{2})$$

Slope:

$$y' = -4 \cos x$$

$$y'(\frac{\pi}{4}) = -4 \cos(\frac{\pi}{4})$$

$$y'(\frac{\pi}{4}) = -4(\frac{1}{\sqrt{2}})$$

$$y'(\frac{\pi}{4}) = -2\sqrt{2}$$

$$m = -2\sqrt{2}$$

Eqⁿ:

$$y = mx + b$$

$$-2\sqrt{2} = -2\sqrt{2}(\frac{\pi}{4}) + b$$

$$-2\sqrt{2} = -\frac{\pi\sqrt{2}}{2} + b$$

$$-2\sqrt{2} + \frac{\pi\sqrt{2}}{2} = b$$

$$y = -2\sqrt{2}x + \frac{\pi\sqrt{2}}{2} - 2\sqrt{2}$$

4) Determine an equation for the tangent to the function $f(x) = \tan x$ at $x = \frac{\pi}{4}$.

Point:

$$f\left(\frac{\pi}{4}\right) = \tan\left(\frac{\pi}{4}\right)$$

$$f\left(\frac{\pi}{4}\right) = 1$$

$$\left(\frac{\pi}{4}, 1\right)$$

Slope:

$$f'(x) = \sec^2 x$$

$$f'\left(\frac{\pi}{4}\right) = \sec^2\left(\frac{\pi}{4}\right)$$

$$f'\left(\frac{\pi}{4}\right) = \frac{1}{\cos^2\left(\frac{\pi}{4}\right)}$$

$$f'\left(\frac{\pi}{4}\right) = \frac{1}{\left(\frac{1}{\sqrt{2}}\right)^2}$$

$$m = 2$$

Eqⁿ:

$$y = mx + b$$

$$1 = 2\left(\frac{\pi}{4}\right) + b$$

$$1 - \frac{\pi}{2} = b$$

$$y = 2x + 1 - \frac{\pi}{2}$$

5) Find an equation of a line that is tangent to $y = 2 \sin x$ and whose slope is a max value.

$y = 2 \sin x$ has a max slope
when $y' = 2 \cos x$ is a max.

$$2 = 2 \cos x$$

$$1 = \cos x$$

$$x = 0, 2\pi, 4\pi, 6\pi, \dots$$

max when $x = 2k\pi, k \in \mathbb{Z}$

(use any answer)

Point:

$$y(0) = 2 \sin(0)$$

$$y(0) = 0$$

$$(0, 0)$$

Slope:

$$y'(0) = 2 \cos(0)$$

$$y'(0) = 2(1)$$

$$y'(0) = 2$$

Eqⁿ:

$$y = mx + b$$

$$0 = 2(0) + b$$

$$b = 0$$

$$y = 2x$$

Answers:

1) a) $\frac{dy}{dx} = 4 \cos x$ b) $f'(x) = 3 \sin x$ c) $\frac{dy}{dx} = -\sin x - \cos x$ d) $\frac{dy}{dx} = 2x - 3 \cos x$ e) $\frac{dy}{dx} = -\sin x + 7\pi \cos x - 3$

f) $\frac{dy}{dx} = -\frac{\pi}{4} \sin x - \frac{\pi}{3} \cos x$

2) $y = -\frac{\sqrt{3}}{2}x + \frac{\pi\sqrt{3}+3}{6}$

3) $y = -2\sqrt{2}x + \frac{\sqrt{2}\pi}{2} - 2\sqrt{2}$

4) $y = 2x + 1 - \frac{\pi}{2}$

5) $y = 2x$; note: there are an infinite number of solutions. The slope is at a max value at any $x = 2k\pi$ where $k \in \mathbb{Z}$. Depending on which x value you choose, you will get a different y -int.